Priyanka Gawade

Curriculum Vitae

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Education

2025-Present **Postdoctoral Researcher**, University of Rome Tor Vergata, Rome, Italy

2020–2025 **PhD**, IUCAA, Pune, India

2019–2020 **Grad School**, IUCAA, Pune, India

2017–2019 **Integrated MSc-PhD**, IISER, Pune, India

2012–2016 **B.Tech**, Mechanical Engineering, VIT, Pune, India

Research Interests

Astrophysics, Cosmology, Gravitational Lensing, Dark Matter, Primordial black holes, Gamma ray bursts, Microlensing, Strong Lensing and Machine Learning.

Technical Skills

Python, Tensorflow, Keras, CNNs, Autoencoders, LaTeX, Linux and MacOS

Collaborations

LSST, Daksha Science

Talks

July 2022 Gravitational Lensing probes of Dark Matter, IIT Bombay, India.

December 2022 Constraining the nature of Dark Matter using Gravitational Lensing Parallax, IAGRG conference, IISER Kolkata, India.

September 2023 Neural network prediction of model parameters for strong lenses from HSC, LSST- joint DESC - SLSC monthly telecon.

December 2023 Hunting primordial black holes using lensing parallax, ICGC, IIT Guwahati, India

March 2024 On the feasibility of primordial black holes-dark matter abundance constraints using microlensing parallax of Gamma ray bursts, Multi-messenger gravitational lensing: Royal society meeting, Manchester, UK

March 2024 Neural network prediction of model parameters of strong lenses from HSC, Rubin LSST strong lensing meeting, University of Oxford, UK

March 2024 Gravitational lensing probes of Dark matter, University of Nottingham, UK

July 2024 Neural network prediction of model parameters for strong lensing samples from Hyper Suprime-Cam Survey, Rubin Community Workshop 2024

January 2025 Neural network prediction of model parameters for strong lenses from HSC, AI/ML in astronomy and astrophysics workshop, IUCAA, India

September 2025 Bayesian Neural network prediction of lens and source parameters along with their uncertainties for strong lenses from HSC, LSST@Europe7

Publications

On the feasibility of primordial black hole abundance constraints using lensing parallax of GRBs. (Gawade et al. 2023, 10.1093/mnras/stad3336)

Neural network prediction of model parameters for strong lensing samples from Hyper Suprime-Cam Survey. (Gawade et al. 2025, 10.1093/mnras/staf935)

Bayesian neural network prediction of lens and source parameters for strong lensing samples from Hyper Suprime-Cam Survey. (Gawade et al. 2026, under review in MNRAS)

Science with the Daksha High Energy Transients Mission. (Bhalerao, Sawant, ... , **Gawade** et al. 2024, Exp Astron 57, 23. https://doi.org/10.1007/s10686-024-09923-1)

Tutoring

Assisted for tutorials and grading assignments during a Cosmology course (Spring 2022) taught by Dr. Surhud More for M.Sc students from University of Pune, India.

IUCAA SciPOP/Public Outreach

Speaker (Public Talk) : Introduction to Gravitational Lensing, Science day celebration, Feb 2020, IUCAA, Pune, India.

Demonstration of a Gravitational Lensing experiment (Group activity), Science day celebration, Feb 2020, IUCAA, Pune, India.

Audio recording for a online presentation (Group activity), Science day celebration, Feb 2022, IUCAA, Pune, India.

Poster presentation (Group activity), The Universe at large scale : visible and invisible, Science day celebration, Feb 2023, IUCAA, Pune, India.

Poster presentation (Group activity), Research in cosmology at IUCAA, Science day celebration, Feb 2024, IUCAA, Pune, India.

Languages

English, Hindi and Marathi

Hobbies

Yoga, Dance, Music, Working towards a healthy and sustainable lifestyle.